

Bayesian Optimization of Multi-Material 3D-Printed Tensegrity Structures for Energy Absorption

BYU Ira A. Fulton College of Engineering
Mentored Research Grant Proposal

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Duration: 2 Years — **Requested:** \$25,000

Abstract

We propose a two-year mentored research program in which 2–3 undergraduate students design and optimize multi-material 3D-printed tensegrity-inspired structures for energy absorption. Tensegrity-inspired structures—assemblies of rigid compression members (PLA) and flexible tension members (TPU)—offer tunable mechanical responses and high strength-to-weight ratios. Using a multifidelity Bayesian optimization framework, students will combine finite element simulations (low-fidelity) with physical experiments (high-fidelity) to efficiently navigate the high-dimensional design space. Students will gain hands-on experience in CAD design, multi-material 3D printing, impact testing, computational modeling, and data-driven optimization. Expected outcomes include undergraduate conference presentations (UCUR, ASME IDETC), a peer-reviewed journal submission, and development of marketable technical skills. This project provides a rich mentored learning environment at the intersection of structural mechanics, advanced manufacturing, and machine learning.

Number of undergraduates: 2–3
Number of graduate students: 1 (co-mentor)

Budget Summary

Category	Year 1	Year 2	Total
Undergraduate wages	\$7,500	\$7,500	\$15,000
Graduate student wages	\$2,000	\$2,000	\$4,000
Supplies	\$2,750	\$750	\$3,500
Travel	\$0	\$2,000	\$2,000
Other	\$250	\$250	\$500
Total	\$12,500	\$12,500	\$25,000

Relationship to External Funding

The PIs do not currently hold external funding for tensegrity or architected materials research. This MRG provides the primary support for establishing a mentored undergraduate research program in this area. If the program proves successful, the preliminary data and mentoring framework developed here may inform a future NSF proposal; however, the MRG is self-contained and its outcomes—trained student researchers, conference presentations, and a journal submission—do not depend on external funding.

1 Research Motivation and Overview

We propose a two-year mentored research program in which 2–3 undergraduate students design and optimize **multi-material 3D-printed tensegrity-inspired structures** for energy absorption. Tensegrity (tensional integrity) structures—assemblies of rigid compression members suspended within a continuous network of tension members—exhibit remarkable strength-to-weight ratios and tunable mechanical responses, making them attractive for protective equipment, packaging, and aerospace applications [Skelton and de Oliveira, 2009, Fraternali et al., 2015]. Recent work has shown that *tensegrity-inspired* architectures can be directly 3D-printed and retain desirable load-limiting behavior under impact [Pajunen et al., 2019]. Because multimaterial FDM TPU elements are not ideal cables, we target tensegrity-inspired architectures that reproduce key tensegrity-like mechanical responses while remaining manufacturable; notably, TPU’s rate-dependent damping is an asset for energy absorption rather than merely a deviation from idealized behavior.

Recent advances in multi-material 3D printing enable the fabrication of tensegrity-inspired structures on a single print platform, using PLA for rigid struts and TPU for flexible tension elements [Ye et al., 2023, Khatri and Egan, 2024]. This eliminates tedious manual assembly and opens a rich, high-dimensional design space. Navigating this space efficiently requires intelligent experimental design; **multifidelity Bayesian optimization** (BO) provides exactly this capability by combining inexpensive simulations (low-fidelity) with physical experiments (high-fidelity) to guide the search toward optimal configurations [Mo et al., 2023, Perdikaris et al., 2017].

The primary objective of this project is to provide a rich, interdisciplinary mentored research experience for undergraduates. Students will gain hands-on ~~experience in CAD design skills in CAD~~, multi-material 3D printing, impact testing, computational modeling, and data-driven ~~optimization—skills directly transferable to graduate school and engineering careers~~optimization. Each student will lead a ~~clearly~~-scoped research project, present at conferences, and ~~contribute to a peer-reviewed co-author a~~ publication.

Figure placeholder: Overview of project approach—tensegrity unit cell design (left); multi-material 3D-printing of PLA struts and TPU tension elements (center); and multifidelity Bayesian optimization loop connecting simulations to experiments (right).—	Figure placeholder: Tensegrity unit cell design (left); multi-material 3D-printing of PLA struts and TPU tension elements (center); and multifidelity Bayesian optimization loop connecting simulations to experiments (right).—
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Figure 1: Conceptual overview of the proposed research and mentoring framework.

2 Background

Tensegrity structures consist of isolated compression members (~~bars or~~ struts) held in ~~stable~~ equilibrium by a continuous network of tension members (~~cables or strings~~) [Skelton and de Oliveira, 2009, Sultan, 2009]. Their defining ~~characteristic—that no~~ characteristic—no two rigid bars touch—yields lightweight assemblies with surprising load-bearing capacity, ~~resilience~~, and tunable non-linear responses [Amendola et al., 2014, Zhang et al., 2015]. Pajunen et al. [Pajunen et al., 2019] demonstrated that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact, ~~confirming their potential for energy absorption without requiring manual cable assembly. Prior work on multi-material.~~ Multi-material rigid-soft printing [Ye et al., 2023, Khatri and Egan, 2024] has shown that PLA-TPU combinations printed

on a single FDM platform ~~can achieve tunable energy absorption, though interface durability under repeated impact remains an active area of investigation.~~ Ye et al. demonstrated a wrapping-based strategy (rigid cores encapsulated by continuous soft skins) that prevents interface delamination and enables cyclic durability—a critical enabler for our fabrication approach.

Bayesian optimization (BO) is a sequential, model-based strategy for optimizing ~~expensive-to-evaluate expensive~~ black-box functions ~~[Shahriari et al., 2016, Frazier, 2018]. A using a~~ Gaussian process surrogate ~~model is fitted to observed data, and an acquisition function balances exploration with exploitation~~ [Shahriari et al., 2016, Frazier, 2018]. BO has been applied to materials discovery [Lookman et al., 2019] and metamaterial design [Wang et al., 2022]. ~~Critically,~~ Mo et al. [Mo et al., 2023] ~~recently demonstrated~~ **multifidelity BO** for architected materials, using ~~inexpensive~~ simulations as a low-fidelity source and ~~physical~~ experiments as high-fidelity data. We adopt this framework ~~directly: finite element: FEA~~ simulations serve as the low-fidelity source, and 3D-printed tensegrity impact tests provide high-fidelity validation. ~~An early calibration campaign will verify low-fidelity high-fidelity correlation before full optimization begins, since~~ Because MFBO effectiveness depends on adequate correlation between fidelity levels, we establish a Year 1 Go/No-Go decision gate: students will validate simulation-experiment correlation on a small calibration set before launching full optimization.

3 Student Research Project 1: Simulation and Bayesian Optimization

Student profile: One junior or senior undergraduate with coursework in mechanics of materials or computational methods.

Scope and Deliverables

This student will develop and run the computational pipeline:

1. **Parameterize a tensegrity unit cell** suitable for simulation and multi-material 3D printing, including strut geometry (PLA), tension element geometry (TPU), topology, and boundary conditions.
2. **Build finite element models** using ABAQUS or open-source FEA tools (e.g., FEniCS) to predict force-displacement response and energy absorption under compaction and drop-test loading. These simulations serve as the *low-fidelity* source in the multifidelity framework. ~~An initial calibration set will validate simulation-experiment correlation before full optimization begins~~
3. **Validate LF-HF correlation (Go/No-Go gate):** Before launching full optimization, the student will run a small calibration set comparing simulated force-displacement curves against experimental results on single unit cells. If the correlation (R^2) between simulated and measured energy absorption is < 0.8 , the team will pivot to single-fidelity experimental BO, ensuring feasibility regardless of model accuracy.
4. **Implement a multifidelity Bayesian optimization loop** (following Mo et al. [Mo et al., 2023]) using Python libraries (BoTorch/Ax) to efficiently explore the design space, fusing simulation predictions with experimental results from Student 2.
5. **Identify optimal designs** and generate predictions for experimental validation.

Mentoring outcomes: The student will develop skills in finite element analysis, scientific computing (Python), and data-driven optimization. They will present results at **ASME IDETC** or **UCUR** and contribute as co-author on a journal manuscript.

4 Student Research Project 2: Fabrication, CAD, and Experimental Testing

Student profile: One junior or senior undergraduate with coursework in machine design, CAD, or compliant mechanisms.

Scope and Deliverables

This student will lead fabrication and physical testing:

1. **Design tensegrity-inspired structures in CAD** (SolidWorks or Fusion 360) for multi-material 3D printing, with PLA struts and TPU tension elements printed on a single platform; ~~incorporating interface design strategies (e.g., interlocking geometries) to ensure durable PLA-TPU bonds [Ye et al., 2023].~~ Following Ye et al. [2023], we adopt a core-wrapping strategy where rigid PLA struts are encapsulated by continuous TPU skins, ensuring robust mechanical interlocking and preventing the interface delamination common in simple layered deposition.
2. **Fabricate structures** using the department's multi-material 3D printer, iterating on print parameters to achieve reliable tensegrity geometries.
3. **Conduct experimental testing:** compaction tests (quasi-static compression) ~~and~~ drop tests (dynamic impact) ~~and wave propagation measurements (hitting on one end and measuring time response on the other)~~ following the drop-weight protocol of Pajunen et al. [2019]. Primary performance metrics are *specific energy absorption* (SEA) and *compaction efficiency* (ratio of absorbed energy to peak force)—tangible, operationally measurable outcomes suitable for undergraduate research. These experiments serve as the *high-fidelity* source.
4. **Compare experimental results** against simulation predictions from Student 1, quantifying model accuracy and informing surrogate updates.

Mentoring outcomes: The student will develop skills in CAD, additive manufacturing, test design, and data analysis. They will present at **UCUR** or **ASME IDETC** and contribute as co-author on a journal manuscript.

5 Mentoring Environment

The mentoring of undergraduate researchers is the central objective of this proposal. We are committed to providing a structured, supportive environment that develops students into independent, capable researchers.

Recruitment

Students will be recruited from courses in mechanics of materials, machine design, kinematics, and compliant mechanisms ~~through brief in-class presentations~~. We target ~~junior or senior undergraduates~~. ~~We will actively seek to recruit students~~ juniors/seniors and will actively recruit from underrepresented groups in engineering.

Mentoring Structure

- **Weekly individual meetings** (30 min) with ~~faculty PI (Prof~~Profs. Hill ~~) and Co-PI (Prof. Baird) and Baird~~ for technical guidance ~~, troubleshooting, and professional development discussions~~.
- **Weekly lab group meetings** (1 ~~hour~~)including hr): progress updates, student ~~presentations on recent literature, career discussions, and literature presentations~~, skill-building workshops (e.g., scientific writing, ~~conference~~ presentation skills, research ethics).
- **Graduate ~~student~~ co-mentor:** Prof. Baird's master's student ~~will serve as~~ provides day-to-day ~~co-mentor, providing hands-on~~ guidance in simulation, coding, ~~fabrication,~~ and experimental technique. This layered ~~mentoring model ensures students receive~~ model ensures timely support while the graduate student develops ~~their own~~ mentoring skills.
- **Progressive responsibility:** Students begin with structured tasks (~~running simulations with provided scripts, assembling structures from detailed CAD plans~~) and progress to independent design decisions, culminating in ownership of their research project.
- **Peer mentoring:** In Year 2, experienced students ~~will~~ mentor newly recruited ~~freshmen or sophomores, reinforcing their own learning while expanding the research group and~~ sophomores, creating a sustainable pipeline of trained researchers.

Student Development Outcomes

~~Students will develop a broad, transferable skill set spanning~~ Technical risk-mitigation activities are structured as learning modules that develop transferable skills:

- **Technical:** experimental mechanics (test design, data acquisition, interface characterization), computation (FEA modeling, Python, Bayesian optimization), and design (CAD, multi-material 3D printing with wrapping strategies).
- **Communication:** lab notebooks, group presentations, conference talks, and co-authoring a peer-reviewed manuscript.
- **Professional:** project management, data-driven decision-making (interpreting Go/No-Go correlation results), collaboration with interdisciplinary teams, and preparation for graduate school or industry careers.

6 Expected Research Outcomes

1. **Trained undergraduate researchers:** 2–3 students with hands-on experience in advanced manufacturing, computational modeling, and data-driven optimization, prepared for graduate study or engineering careers.
2. **Conference presentations:** Student-led presentations at **UCUR** (Utah Conference on Undergraduate Research) and **ASME IDETC** (International Design Engineering Technical

Conferences).

3. **Journal submission:** One peer-reviewed manuscript submitted to the *ASME Journal of Mechanical Design* or *Smart Materials and Structures*, with undergraduate students as co-authors.
4. **Open-source tools:** Simulation code, optimization scripts, and experimental data published on GitHub for reproducibility.

7 Potential Impact of Work

This project advances ~~the design of architected materials~~ architected material design for energy absorption with ~~direct~~ applications in **protective equipment** (helmets, body armor), **packaging** (shock-absorbing inserts), and **aerospace** (lightweight impact-resistant structures). By demonstrating that multifidelity ~~Bayesian optimization BO~~ BO can efficiently navigate the tensegrity design space, we establish a generalizable framework ~~applicable to other classes of architected materials and metamaterials.~~

~~The interdisciplinary training—spanning structural mechanics, advanced manufacturing, and machine learning—prepares for other metamaterial classes.~~ The interdisciplinary training prepares students for graduate study or careers in ~~industries increasingly reliant on~~ data-driven design. ~~The open-source~~ Open-source dissemination of tools and data extends impact beyond BYU ~~and enables other research groups to build on our work.~~

8 Timeline

Task	Y1 Fall	Y1 Winter	Y2 Fall	Y2 Winter
Student recruitment & onboarding	•			
Literature review	•	•		
Tensegrity parameterization & CAD	•	•		
<u>Interface testing & print validation</u>	<u>•</u>	<u>•</u>		
FEA simulation development		•	•	
<u>LF-HF correlation gate (Go/No-Go)</u>		<u>•</u>		
Multifidelity BO implementation		•	•	<u>•</u>
Multi-material 3D printing & fabrication		•	•	•
Impact testing & data collection			•	•
Simulation–experiment comparison				•
Conference presentations (UCUR/IDETC)			•	•
Journal manuscript preparation				•
Peer mentoring of new students			•	•

Table 1: Project timeline across four semesters.

9 Budget

Total requested: **\$25,000** over 2 years. Per grant guidelines, no funds are allocated to equipment, tuition, or faculty wages.

Category	Description	Year 1	Year 2
Undergraduate wages	2–3 students, hourly RA	\$7,500	\$7,500
Graduate student wages	Partial RA for grad co-mentor	\$2,000	\$2,000
Supplies	Filament, load cells, DAQ, compute	\$2,750	\$750
Travel	Conference registration & travel	\$0	\$2,000
Other	Poster printing, consumables	\$250	\$250
Annual Total		\$12,500	\$12,500

Undergraduate wages (\$15,000): Supports 2–3 undergraduates for simulation, 3D printing, and testing. **Graduate student wages (\$4,000):** Prof. Baird’s master’s student serves as day-to-day co-mentor. **Supplies (\$3,500):** PLA/TPU filament, load cells, DAQ module, fasteners, and cloud compute credits (Year 1 heavy). **Travel (\$2,000):** Student presentations at ASME IDETC or UCUR (Year 2). **Other (\$500):** Poster printing, consumables, contingency.

References

- Ada Amendola, Everth Hernandez-Nava, Russell Goodall, Iain Todd, Hugh Shercliff, Hugh Shercliff, and Robert E. Skelton. Experimental investigation of the softening–hardening response of tensegrity prisms under compressive loading. *Composite Structures*, 117:234–243, 2014. doi: 10.1016/j.compstruct.2014.06.025.
- Fernando Fraternali, Gerardo Carpentieri, and Ada Amendola. Tensegrity ring structures for impact protection. *International Journal of Solids and Structures*, 65–66:232–243, 2015. doi: 10.1016/j.ijsolstr.2015.03.024.
- Peter I. Frazier. A tutorial on Bayesian optimization. *arXiv preprint arXiv:1807.02811*, 2018. URL <https://arxiv.org/abs/1807.02811>.
- Nava Raj Khatri and Paul F. Egan. Energy absorption of 3D printed ABS and TPU multimaterial honeycomb structures. *3D Printing and Additive Manufacturing*, 11:e840–e850, 2024. doi: 10.1089/3dp.2022.0196.
- Turab Lookman, Prasanna V. Balachandran, Dezhen Xue, and Ruihao Yuan. Active learning in materials science with emphasis on adaptive sampling using uncertainties for targeted design. *npj Computational Materials*, 5(1):21, 2019. doi: 10.1038/s41524-019-0153-8.
- Changyu Mo, Paris Perdikaris, and Jordan R. Raney. Accelerated design of architected materials with multifidelity Bayesian optimization. *Journal of Engineering Mechanics*, 149(6):04023028, 2023. doi: 10.1061/JENMDT.EMENG-7033.
- Kirsti Pajunen, Paul Johanns, Raj Kumar Pal, Julian J. Rimoli, and Chiara Daraio. Design and impact response of 3D-printable tensegrity-inspired structures. *Materials & Design*, 182:107966, 2019. doi: 10.1016/j.matdes.2019.107966.
- Paris Perdikaris, Maziar Raissi, Andreas Damianou, Neil D. Lawrence, and George Em Karniadakis. Nonlinear information fusion algorithms for data-efficient multi-fidelity modelling. *Proceedings of the Royal Society A*, 473(2198):20160751, 2017. doi: 10.1098/rspa.2016.0751.

- Bobak Shahriari, Kevin Swersky, Ziyu Wang, Ryan P. Adams, and Nando de Freitas. Taking the human out of the loop: A review of Bayesian optimization. *Proceedings of the IEEE*, 104(1): 148–175, 2016. doi: 10.1109/JPROC.2015.2494218.
- Robert E. Skelton and Mauricio C. de Oliveira. *Tensegrity Systems*. Springer, New York, 2009. doi: 10.1007/978-0-387-74242-7.
- Cornel Sultan. Tensegrity: 60 years of art, science, and engineering. *Advances in Applied Mechanics*, 43:69–145, 2009. doi: 10.1016/S0065-2156(09)43002-3.
- Yue Wang et al. Bayesian optimization for the design of mechanical metamaterials. *Journal of the Mechanics and Physics of Solids*, 159:104734, 2022. doi: 10.1016/j.jmps.2021.104734.
- Haitao Ye, Qingjiang Liu, Jianxiang Cheng, Honggeng Li, Bingcong Jian, Rong Wang, Zechu Sun, Yang Lu, and Qi Ge. Multimaterial 3D printed self-locking thick-panel origami metamaterials. *Nature Communications*, 14:1607, 2023. doi: 10.1038/s41467-023-37343-w.
- Li-Yuan Zhang, Yue Li, Yan-Ping Cao, and Xi-Qiao Feng. Tensegrity cell mechanical metamaterial with metal rubber. *Applied Physics Letters*, 106(20):201901, 2015. doi: 10.1063/1.4921402.

Biographical Sketches

Jeffrey R. Hill — Principal Investigator

Professional Preparation

Ph.D.	Mechanical Engineering	University of Utah
M.S.	Mechanical Engineering	Brigham Young University
B.S.	Mechanical Engineering	Brigham Young University

Appointments

- Associate Professor, Department of Mechanical Engineering, Brigham Young University (current)
- Research Engineer, Sandia National Laboratories (prior)

Products Most Closely Related to This Proposal

Publications and products related to smart materials, shape memory alloys, compliant mechanisms, and structural mechanics.

Synergistic Activities

- Mentoring of undergraduate and graduate students in experimental mechanics and smart materials research at BYU.
- Teaching courses in mechanics of materials, machine design, and kinematics in the Department of Mechanical Engineering.
- Collaboration with national laboratories on advanced materials characterization and structural design.

Sterling G. Baird — Co-Principal Investigator

Professional Preparation

Ph.D.	Materials Science and Engineering	University of Utah
B.S.	Materials Science and Engineering	Brigham Young University

Appointments

- Faculty, Department of Mechanical Engineering, Brigham Young University (current)

Products Most Closely Related to This Proposal

Publications and products related to materials informatics, Bayesian optimization, data-driven materials discovery, and machine learning for materials science.

Synergistic Activities

- Development of open-source tools for Bayesian optimization and materials informatics.
- Mentoring of graduate and undergraduate students in data-driven methods for materials science and engineering.
- Active contributions to the materials informatics community through publications, software, and educational resources.